

Java review, ADTs, and ArrayList

3/28/25

(Including green slides based on lecture by Cynthia Lee)

Administrivia

- HW 1 (clicker ID, codingbat, and essay) due Monday and Wednesday
- Recorded class on Monday; no in-person meeting

Which of the following declares
an array of integers?

- A. `int[] nums;`
- B. `integer[] nums;`
- C. `nums = new int[10];`
- D. `nums = new integer[10];`
- E. Not exactly one of the above

Which of the following declares
an array of integers?

A. int[] nums;

B. integer[] nums;

C. nums = new int[10];

D. nums = new integer[10];

E. Not exactly one of the above

Which of the following is the signature for a method that takes a Person and returns a Dwelling?

- A. `public Dwelling(Person resident)`
- B. `public Dwelling getResidence(Person p)`
- C. `public Person(Dwelling d)`
- D. `public Person getResident(Dwelling d)`
- E. Not exactly one of the above

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- C. `public Person(Dwelling d)`
- D. `public Person getResident(Dwelling d)`
- E. Not exactly one of the above

Which of the following correctly replaces the contents of the array `nums` with all 0s?

- A.

```
for(int i=0; i<10; i++)  
    nums[i] = 0;
```
- B.

```
for(int i=0; i<nums.size; i++)  
    nums[i] = 0;
```
- C.

```
for(int i=0; i<nums.length(); i++)  
    nums[i] = 0;
```
- D.

```
int i = 0;  
while(i < nums.length)  
    nums[i] = 0;
```
- E. Not exactly one of the above

Which of the following correctly replaces the contents of the array `nums` with all 0s?

A. `for(int i=0; i<10; i++)`

`nums[i] = 0;`

B. `for(int i=0; i<nums.size; i++)`

`nums[i] = 0;`

C. `for(int i=0; i<nums.length(); i++)`

`nums[i] = 0;`

D. `int i = 0;`

`while(i < nums.length)`

`nums[i] = 0;`

E. Not exactly one of the above (none are correct)

Which of the following returns whether integer x is between 5 and 10 (inclusive)?

- A. `if((x <= 5) || (x >= 10))
 return true;`
- B. `if((x >= 5) && (x <= 10))
 return true;`
- C. `return ((x >= 5) && (x <= 10));`
- D. `return !((x < 5) || (x > 10));`
- E. Not exactly one of the above

Which of the following returns whether integer x is between 5 and 10 (inclusive)?

A. `if(( $x \leq 5$ ) || ( $x \geq 10$ ))`

`return true;`

B. `if(( $x \geq 5$ ) && ( $x \leq 10$ ))`

`return true;`

C. `return (( $x \geq 5$ ) && ( $x \leq 10$ ));`

D. `return !(( $x < 5$ ) || ( $x > 10$ ));`

E. Not exactly one of the above (C & D)

Abstract data types (ADTs)

- Set of operations that are supported on a particular kind of data
 - Example: List is ADT for storing ordered collection of objects. Supports operations to add objects, remove them, count them, test for a particular object, ...
- Does not include implementation details
 - Language independent
 - May have different implementations that optimize different operations

Which of the following is part of the List ADT (stores ordered sequence)?

- A. An attribute num storing the number of values in the List
- B. A size method that returns the number of values in the List
- C. The size of the array storing values that are put into the List
- D. The main function that creates a List and tests it by calling the methods
- E. Not exactly one of the above

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- D. The main function that creates a List and tests it by calling the methods
- E. Not exactly one of the above

List ADT

- Store ordered collection of values
 - void add(item) //add to end
 - void add(int, item) //add at a given position
 - int size()
 - void remove(item)
 - item remove(int) //remove from given position
 - void clear()
 - item set(int, item)
 - item get(int)
 - ...

What type of item should a List contain?

- List of Strings
- List of ints
- List of Students
- ...

Generics: Generalizing the type

- Allow a single List declaration to work for lists of different kinds of items

```
public class OurArrayList<T > {  
    public void add(T item) { ... }  
  
    public T get(int i) { ... }  
}
```


List ADT

- Store ordered collection of values
 - void add(T) //add to end
 - void add(int, T) //add at a given position
 - int size()
 - void remove(T)
 - T remove(int) //remove from given position
 - void clear()
 - T set(int, T)
 - T get(int)
 - ...

Messy details about generics

- T must be an object type (not “int” or other primitive type)
- Tricky syntax to create an array of this type:

```
@SuppressWarnings("unchecked")  
public OurArrayList() {  
    num = 0;  
    vals = (T[]) new Object[3];  
}
```
- Also, toArray needs to return Object[]